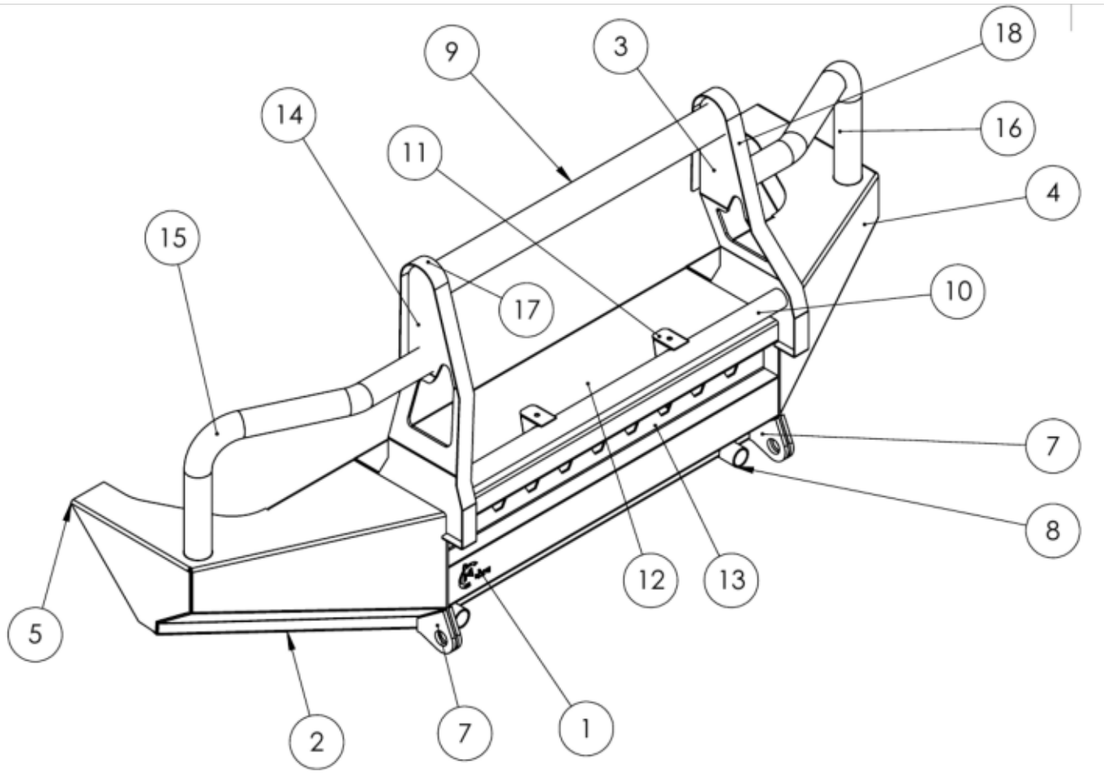
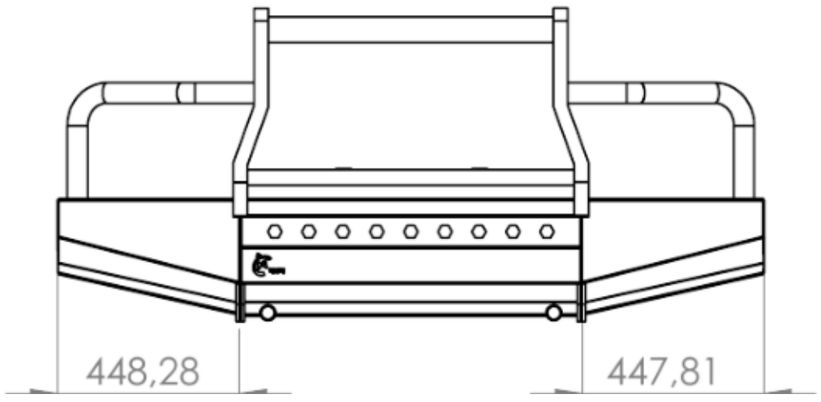
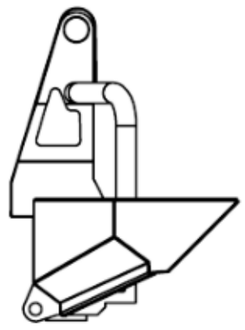
ASSEMBLY - item balloons per source BOM (see table)



FRONT / SIDE - overall 1741 mm; centre 835 mm; ends 448 mm





BILL OF MATERIALS / CUT LIST

Item	VA part no.	Description	Thk	Qty	Flat (mm)	kg ea
1	VA-BB-01	Centre panel	3	1	835 × 209	4.05
2	VA-BB-02	End panel (item 2)	3	1	688 × 687	6.32
3	VA-BB-03	Upright RH	5	1	306 × 769	4.66
4	VA-BB-04	End panel, opposite hand (item 4)	3	1	688 × 687	6.32
5	VA-BB-05	End plate	3	2	269 × 78	0.50
6	VA-BB-06	Bracket 1 (chassis bracket)	5	2	184 × 258	1.82
7	VA-BB-07	Recovery lug	12	4	91 × 92	0.44
8	VA-BB-08	Pipe lift (jacking stub)	38.1 × 2.0	2	L = 100	0.18
9	VA-BB-09	Pipe centre (top hoop)	63.5 × 2.0	1	L = 702	2.15
10	VA-BB-10	Pipe spot (light bar)	38.1 × 2.0	1	L = 805	1.45
11	VA-BB-11	Spot bracket (light mount)	2	2	60 × 89	0.06
12	VA-BB-12	Centre panel P2 (top plate)	3	1	835 × 289	5.73
13	VA-BB-13	Centre panel P3 (lower channel)	3	1	835 × 153	2.87
14	VA-BB-14	Upright LH	5	1	306 × 769	4.66
15	VA-BB-15	Tube 50 LH (side hoop)	50 × 2.0	1	L = 510 (bent)	1.22
16	VA-BB-16	Tube 50 RH (side hoop)	50 × 2.0	1	L = 510 (bent)	1.22
17	VA-BB-17	Face 1 (RH trim strip)	2	1	860 × 87	0.48
18	VA-BB-18	Face 1 LH (trim strip)	2	1	860 × 87	0.48

NEST GROUPS BY THICKNESS (laser)

2 mm 304 SS: 11 ×2, 17 ×1, 18 ×1 | net 0.07 m2 | 1.1 kg

3 mm 304 SS: 01 ×1, 02 ×1, 04 ×1, 05 ×2, 12 ×1, 13 ×1 | net 1.11 m2 | 26.3 kg

5 mm 304 SS: 03 ×1, 06 ×2, 14 ×1 | net 0.33 m2 | 13.0 kg

12 mm 304 SS: 07 ×4 | net 0.02 m2 | 1.8 kg

PROCESS NOTES

1. Laser cut all plate 1:1 from DXF (layer CUT). Layers BEND_UP / BEND_DOWN are bend-line centres - do not cut.
2. Layer ETCH = low-power surface mark (wordmark on centre panel). Layer INFO = reference text only.
3. Flats are developed at K = 0.46 (air bend, 304). 304 springs back 2-3 deg more than mild steel - over-bend and check first-off against the bend table.
4. Laser cut with nitrogen (oxide-free edge). Deburr; break sharp corners 0.5 mm. Stainless-only tooling, films and brushes - no carbon-steel contamination. 12 mm lugs: plasma or 6 kW+ fibre laser.
5. Bend direction is viewed from the flat-pattern face shown on each sheet (UP = flange toward viewer).
6. Tube items 8/9/10/15/16: cut lengths and bend data on the TUBE SCHEDULE sheet, measured from the 1:20 assembly views. Wall 2.0 mm on all tubes.
7. Items 17/18 (face strips): bend as drawn (the source "NO BENDING" note is superseded). The R42 wrap matches the upright top radius.
8. Weld sequence: tack brackets (06) to centre (01) and P2/P3 (12/13); fit uprights (03/14) and lugs (07); add ends (02/04) and end plates (05) last.
9. Several source bend radii are below 1T (red warnings on sheets 05, 06, 11, 03/14, 17/18). Workable in 304 with a sharp-nose punch; confirm on first-off - if it cracks, open the radius.

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-ASM

TITLE
BULL BAR - ASSEMBLY, BOM & CUT LIST

MATERIAL **304 SS 2B**	SHEET PARTS **19**	PLATE / TUBE MASS **42.1 / 6.4 kg**
OVERALL WIDTH 1741 mm	CENTRE SECTION 835 mm	END SECTIONS 448 mm each
THICKNESSES 2 / 3 / 5 / 12 mm	FITMENT TBC (per source 4×4 design)	TUBE CHS 50 / 63.5 / 38.1
WELDING TIG/MIG, ER308L, purge	FINISH Brushed, passivated	PACK VA-BB rev A

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